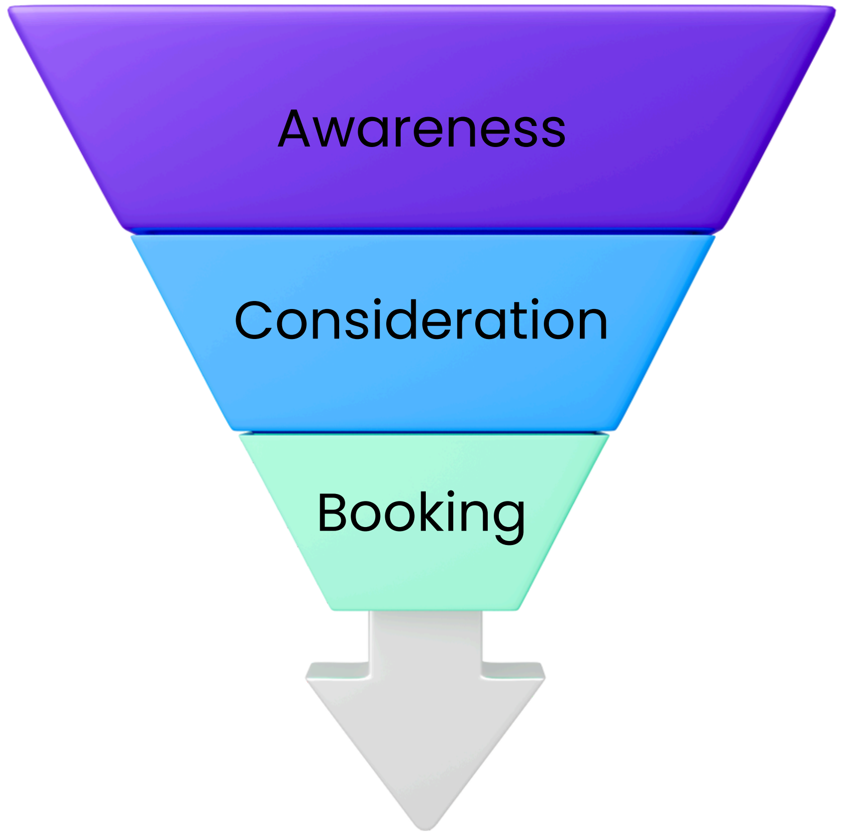
Globe Stay

Travel Project

Presented by
Team 2

MMA 831
3 November 2025

Problem & Motivation



Competition

GlobeStay operates in a highly competitive online-travel space with heavy multi-channel spend and tight ROI pressures.

Lack of Clarity

Lack of clarity on which channels and markets truly drive bookings → inefficient budget allocation.

ROI Optimization

Every €1 reallocated from low- to high-ROI media yields measurable lift and long-term brand equity gains.

Questions & Goals

- **S** Specific – Quantify channel effectiveness across US, UK, DE.
- **M** Measurable – Estimate elasticity & ROI per channel.
- **A** Achievable – Use 190 weeks x 3 countries panel data.
- **R** Relevant – Optimize budget allocation and media mix.
- **T** Time-bound – Produce actionable insights for next fiscal year.

Which channels drive bookings most effectively?

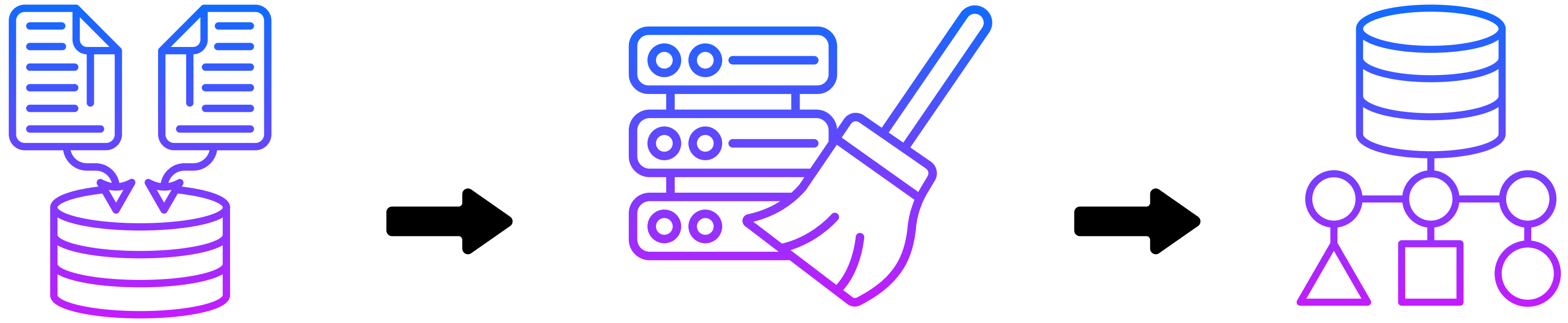
How do performance, awareness, and offline media compare?

Do responses differ across countries?

How do seasonality and external factors affect demand?

How can the mix be optimized for ROI and growth?

Data Preparation



Raw Data → Clean Data → Processed Data

Data Overview

Marketing and bookings data across the US, UK, and Germany (~190 weeks) with weekly spend, impressions, and conversions by channel and country; added trend, seasonality, and currency controls.

See Appendix for more details.

Sources

Weekly spend, impressions, and bookings across 3 countries + trend, seasonality, currency rate.

Cleaning

Missing values handled, unified currency (EUR), weekly aggregation.

Feature Engineering

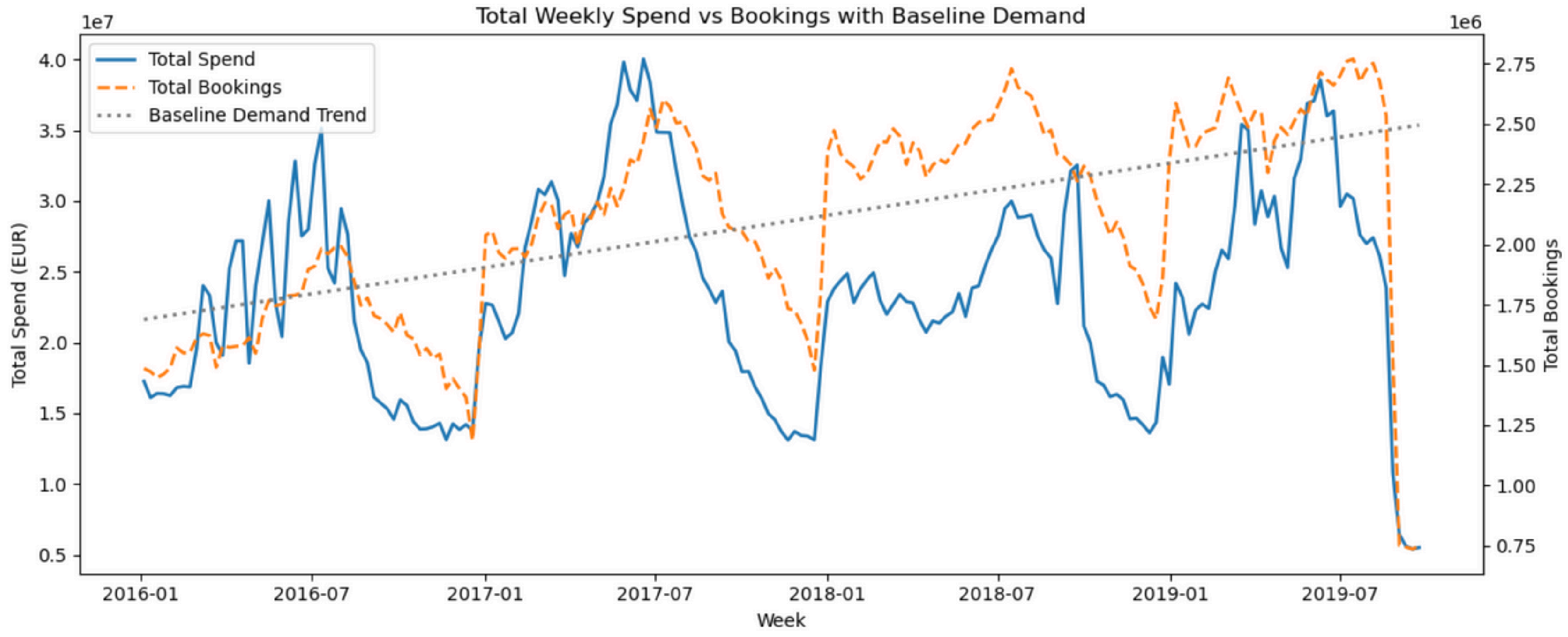
- Trend + sin/cos seasonality controls
- Adstock (lag/decay $\alpha \approx 0.6$)
- Log-scaling for diminishing returns
- Country & channel dummies
- Control variables for holidays and competition

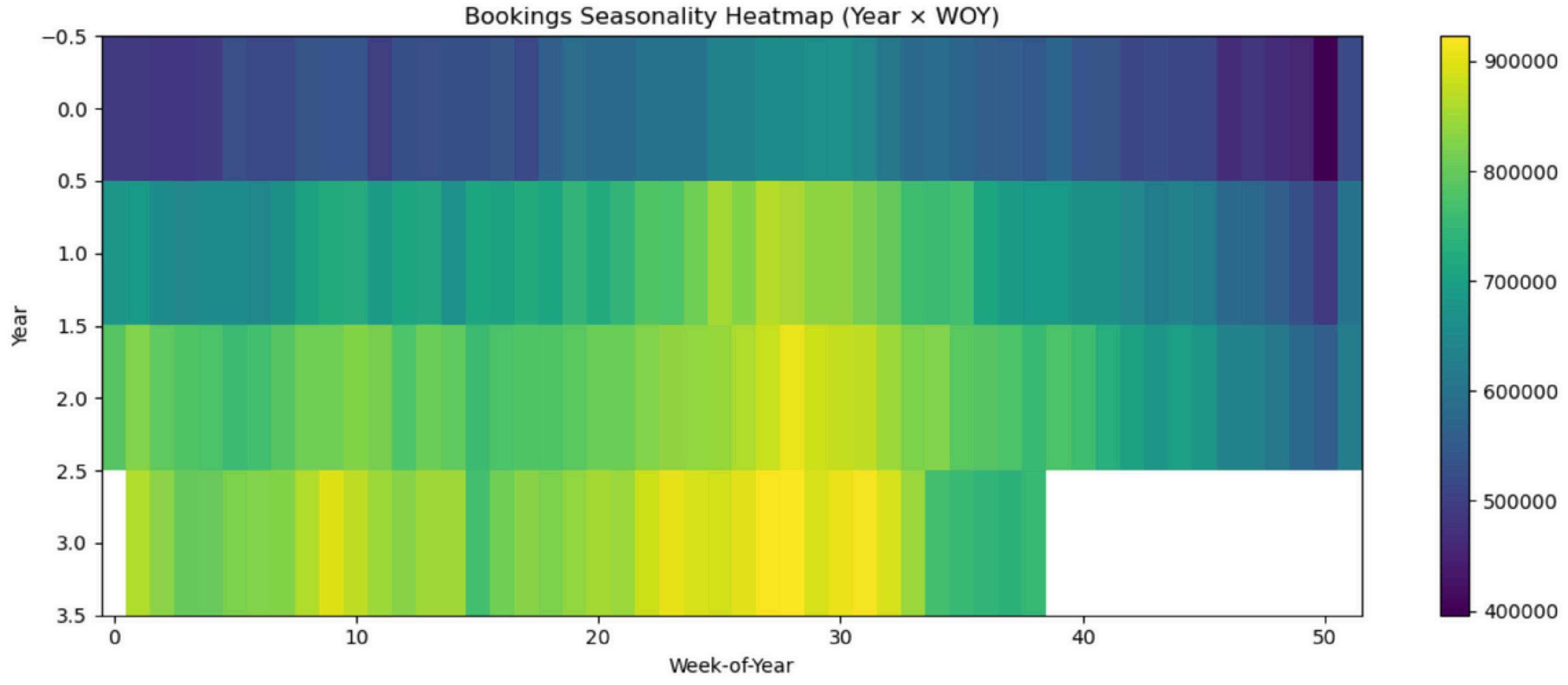
Overall Trends

Insights

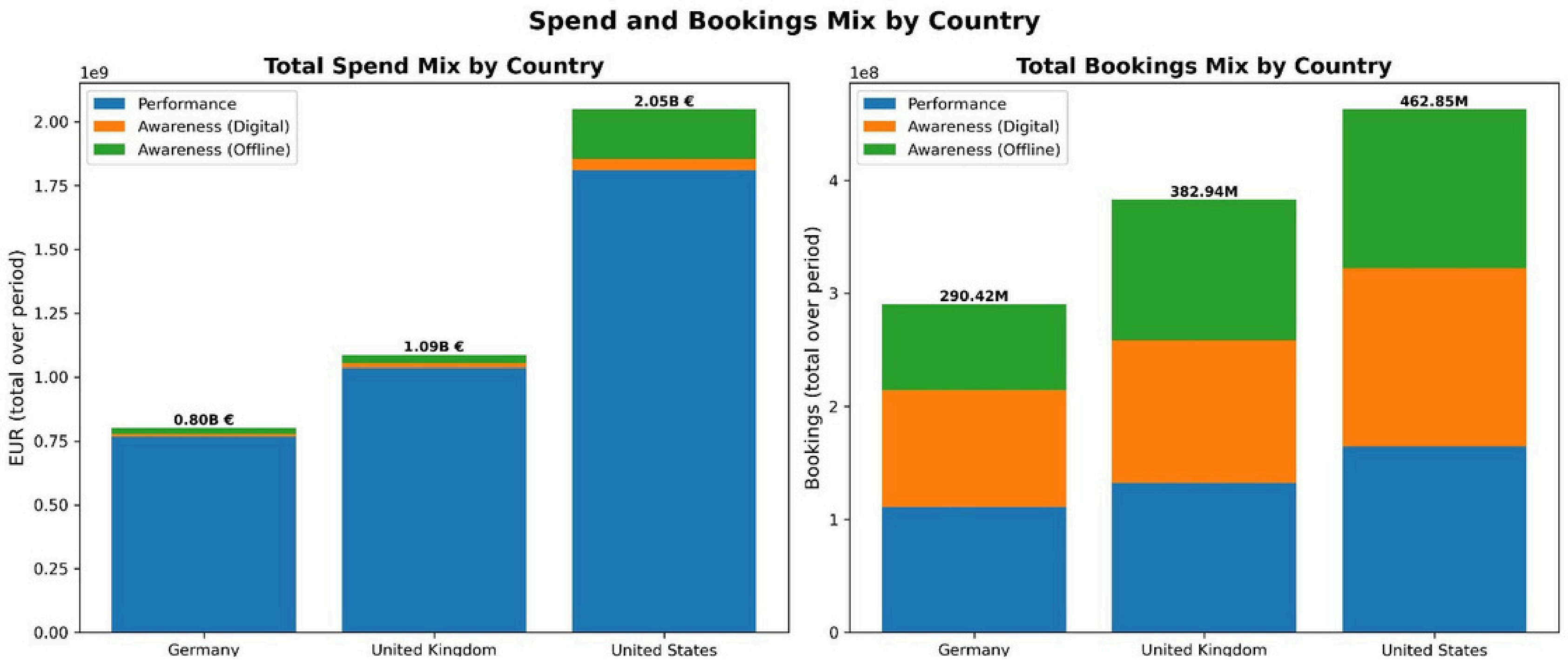
- Spend cycles mirror seasonal peaks.
- Bookings follow a lagged pattern → potential spend inefficiency.
- Mid-2018 shows weaker spend-to-booking response, indicating possible marketing inefficiency
- Baseline demand trending upward.
- Bookings peak consistently in summer - clear seasonal demand cycles.

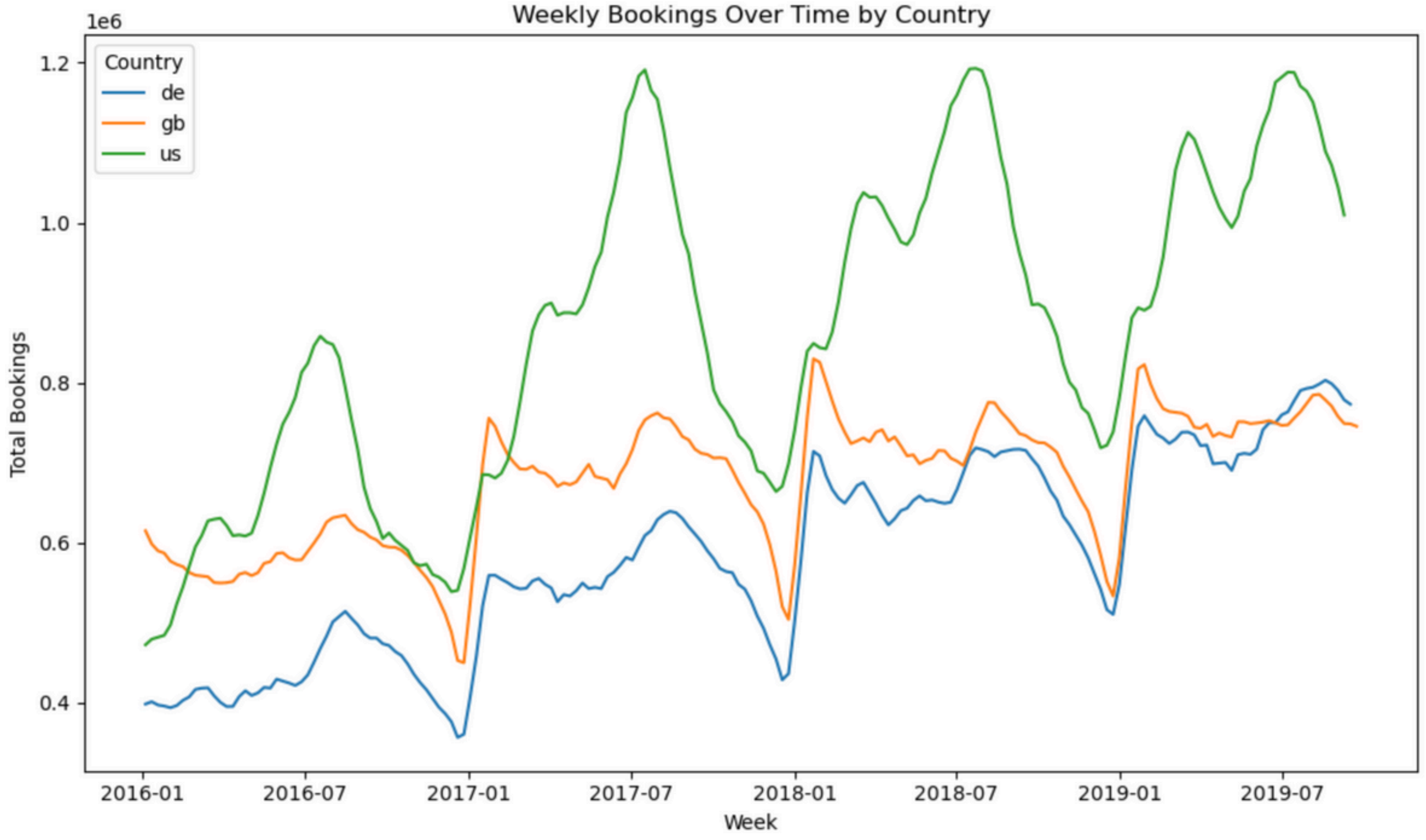
See Appendix for detailed trends by country.





Country-Level Data





Insights

- Spend is heavily skewed toward performance, yet bookings are more balanced, showing strong indirect impact from awareness campaigns.
- Germany shows higher efficiency (bookings/spend), while the US drives volume but shows signs of performance saturation.
- All markets peak in summer and dip in winter.
- US shows the strongest volatility, indicating high elasticity during peak periods.

Model Selection

Model Type

Adstocked, log-log OLS with AR(1) correction (HAC-robust).

Justification

Captures lag effects, diminishing returns, autocorrelation.

Variables

Performance & awareness spend, trend, seasonality, currency, country dummies + interactions.

Diagnostics

Using **Adj R^2, DW, VIF, Test MAPE** to validate the model.

Key Insights

- Captures both short- and long-term media effects.
- Balances realism with clear, actionable outputs.
- Stable across countries and channel types.
- Controls for trend, seasonality, and overlap.
- Supports data-driven budget optimization.

Model Validation

Adj R^2	0.666	Strong explanatory power, minimal overfitting.
DW	1.75	Residuals show no serious autocorrelation.
MAPE	≈ 10.4 %	Excellent predictive accuracy for weekly data.
VIF	1.2 – 6.6	No multicollinearity issues (below 10).
p-values	≈ 85 % < 0.05	Statistically significant coefficients.

See Appendix for full model diagnostics.

01 Statistically Stable

Explains ~67 % of weekly booking variation (Adj R^2 = 0.666) - robust fit for real-world marketing data.

02 Explanatory Power

Most variables are significant - p-values, VIF, and DW confirm reliable, unbiased estimates.

03 Predictive Accuracy

MAPE ≈ 10% means predicted bookings track actual results closely - strong foundation for ROI and scenario simulations.

Channel Effectiveness Overview

Group	Elasticity	ROI	Comments
Performance	0.70	High	Direct response driver
Awareness	0.009	Low	Long-term halo

Elasticities estimated globally from the core model (Performance vs. Awareness); see Appendix for detailed results and supporting data.

Performance channels drive short-term growth

PPC, Metasearch generate the strongest immediate returns - every +10% in spend drives roughly +7% more bookings.

Awareness channels build long-term equity, not instant ROI

Low elasticity shows limited short-run impact - awareness value lies in sustaining brand visibility and fueling future Direct demand.

Offline media provide steady brand reinforcement

Offline spend (TV, Radio, and Print) is included within the Awareness category and supports brand-building efforts rather than driving direct conversions.

Country-Level Effectiveness

Country	Responsiveness	Key Insights
US	High (Strong signal)	Highest mean f_perf. Bookings and spend move in sync - performance media scales efficiently.
Germany	Moderate (Steady/Seasonal)	Lower f_perf mean. Clear seasonality; awareness sustains baseline demand.
UK	Low/Irregular	Weak, inconsistent f_perf signal. Possible data or creative inefficiency.

See Appendix for detailed results and supporting data.

Recommended Action

US

Strongest Responsiveness

Maintain performance-heavy mix; optimize bids and timing for peak ROI.

Germany

Steady, Seasonal, Less Elastic

Align spend with seasonal peaks; use awareness for pre-season lift.

UK

Weak/Irregular Response

Audit data; test new creatives and audience mixes; rebalance spend.

Attribution & Scenario Analysis

+10% Performance

Drives roughly 7% more bookings, confirming these channels' high ROI.

+10% Offline Spend

Limited short-run ROI; brand reinforcement rather than direct performance impact.

+10% Awareness

Minimal immediate response; primarily brand-building rather than short-term conversion.

TV → PPC

Improves total bookings by ~0.06%; smarter mix optimization beats incremental spending.

Scenario	Result
+10 % Performance	+7.4 % bookings (high ROI)
+10 % Awareness	+0.1 % (brand-only effect)
+10% Offline Spend	Minimal immediate effect
TV → PPC shift	0.06 % (neutral)

See Appendix for detailed results and supporting data.

Strategic Recommendations

Performance Spend	Awareness Strategy	Offline Media
Increase	**Maintain**	**Trim**
Increase investment by 6% in high-response markets (US, DE) where marketing delivers strong short-term ROI. → **Expected impact:** ~+0.4% incremental bookings within efficient ROI range.	**Allocate 8% of total spend** to awareness to preserve brand strength, focusing on high-performing digital channels. → **Long-term role:** sustain baseline demand rather than drive immediate lift.	**Trim 5% from low-performing offline** placements and reinvest into performance or digital awareness. → **Scenario tests show:** <0.1% impact on bookings with improved overall ROI efficiency.

Conclusion

Business Impact

Bookings Increase	Annual Growth of Organic Demand	ROI Improvement
+1%	**+3%**	**+8 %**

Reallocating 5% from low-performing offline media to **high-response digital** channels can lift bookings, sustain organic demand growth, and improve ROI, resulting in ≈ **€4 million** in annual savings.

If GlobeStay expands its budget, it should invest in **high-response performance channels** in the **US** and **Germany**, markets with the strongest booking returns and ROI scalability.

Thank You

GlobeStay Project

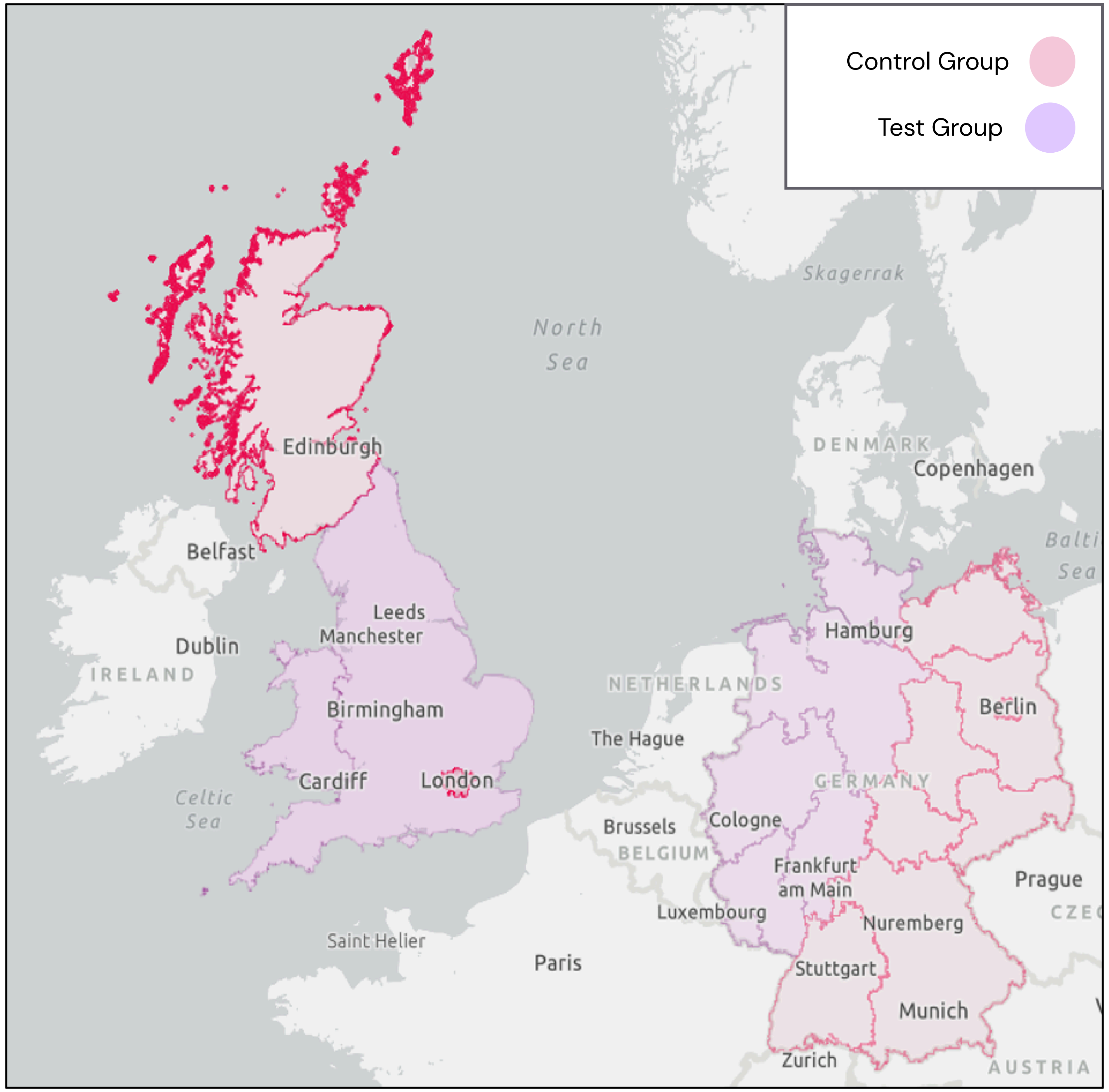
Team 2

Appendix

Sample Experiment

TV → PPC Scenario

- **Design:** 10% TV reallocated to Branded PPC
- **Duration:** 8 weeks
- **Control Groups:** Geographical split for UK and Germany
- **Goal:** Improve cost per booking
- **Metrics:** Bookings lift, CPB, brand search volume
- **Next Step:** Scale if ROI improves >=10% with no drop in brand demand



Data Preparation

Steps

1. Aggregated to weekly level by country and channel.
2. Missing values: short gaps forward-filled; longer ones interpolated.
3. Currency normalization: all costs converted to EUR (USD, GBP → EUR).
4. Outlier treatment: weekly spend/bookings winsorized at $\pm 3\sigma$.
5. Temporal check: verified continuous weekly timeline per country.
6. Control variables: holidays, seasonality, and competitive pressure modelled via feature transformations.

Feature Engineering

Feature Type	Transformation	Purpose
Trend	Linear & log trend	Capture baseline growth in bookings.
Seasonality	$\sin(2\pi t/52)$ and $\cos(2\pi t/52)$	Model yearly demand cycles.
Adstock	Lag decay ($\alpha = 0.6$)	Reflect carryover effects of media exposure.
Log Scaling	$\log(1 + \text{spend})$	Handle diminishing returns.
Controls	Holiday dummy, Competitive index	Adjust for exogenous shocks.
Country dummies	US, UK, DE	Capture market-level differences.

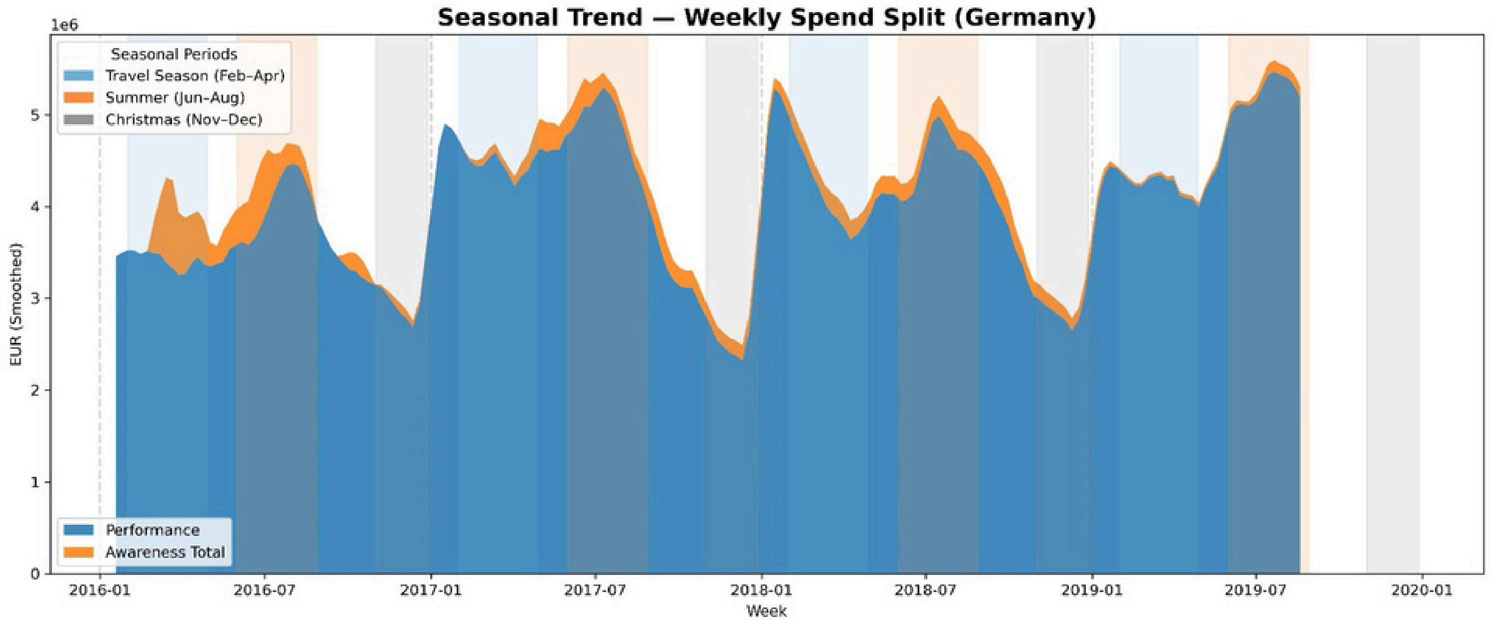
Appendix

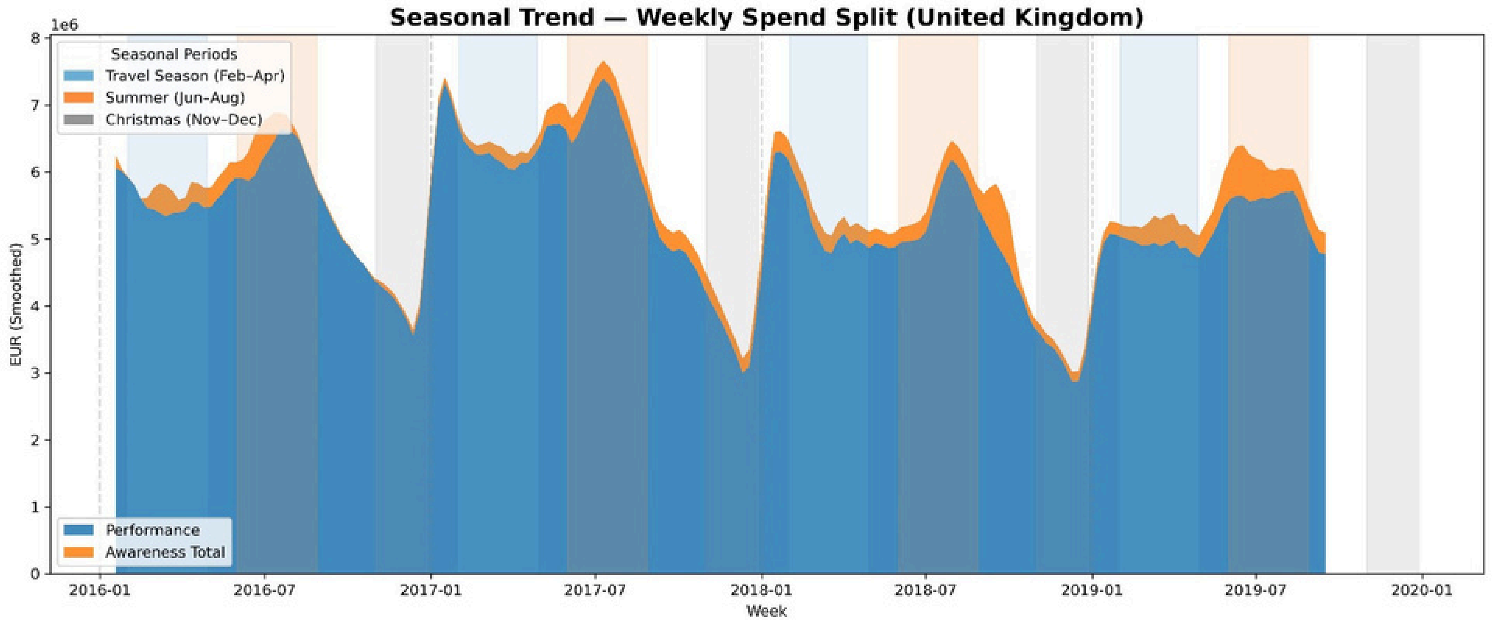
Country-Level Data

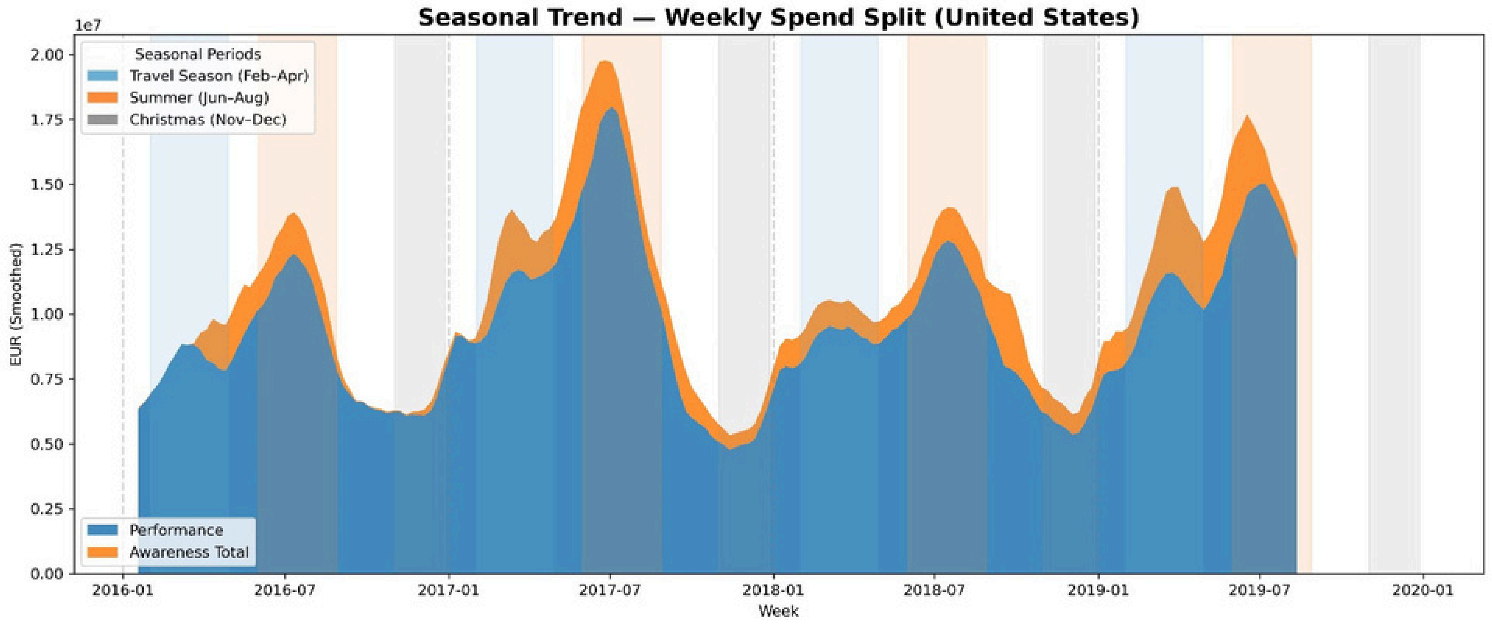
United States:
Performance spend dominates and scales strongly with travel and summer peaks (June–August). Clear cyclicality shows efficient alignment with booking demand, suggesting strong ROI seasonality.

United Kingdom:
Spending follows a similar pattern but with flatter amplitude, indicating steadier investment across periods and lower elasticity to seasonal peaks. Awareness spend remains proportionally stable.

Germany:
Spend patterns mirror broader seasonality, with summer and Christmas peaks, but show sharper post-season declines. Suggests opportunity to re-time awareness investment to sustain momentum after peak travel months.







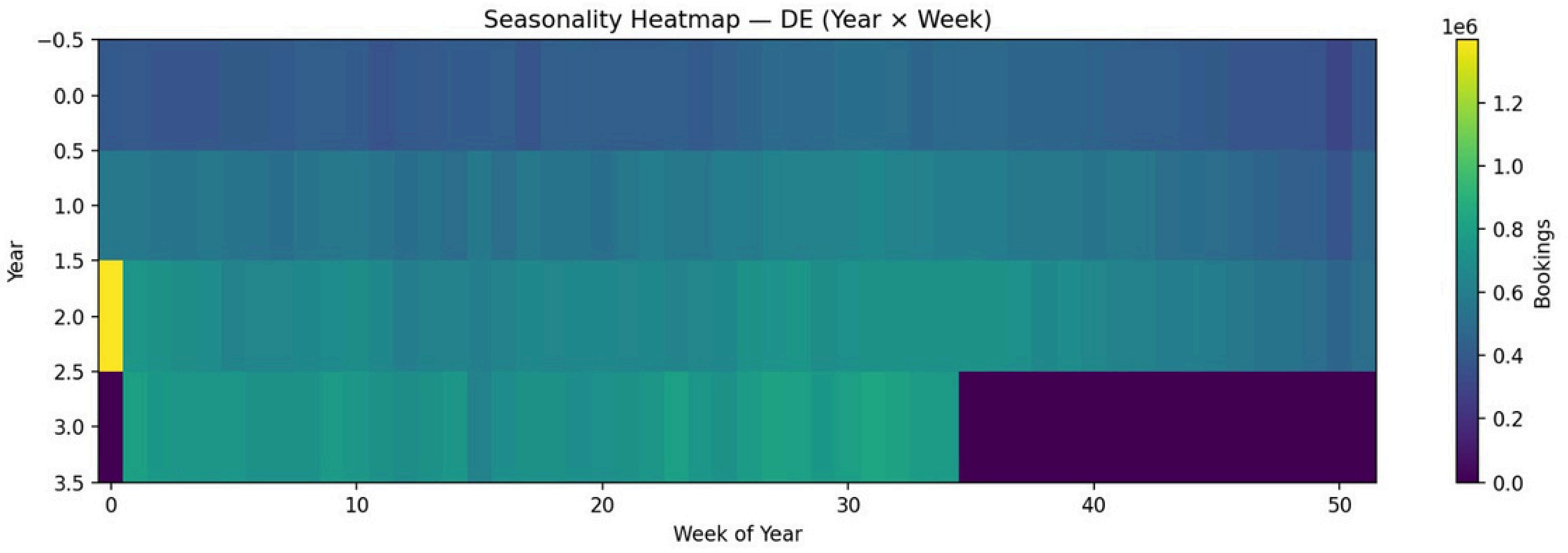
Appendix

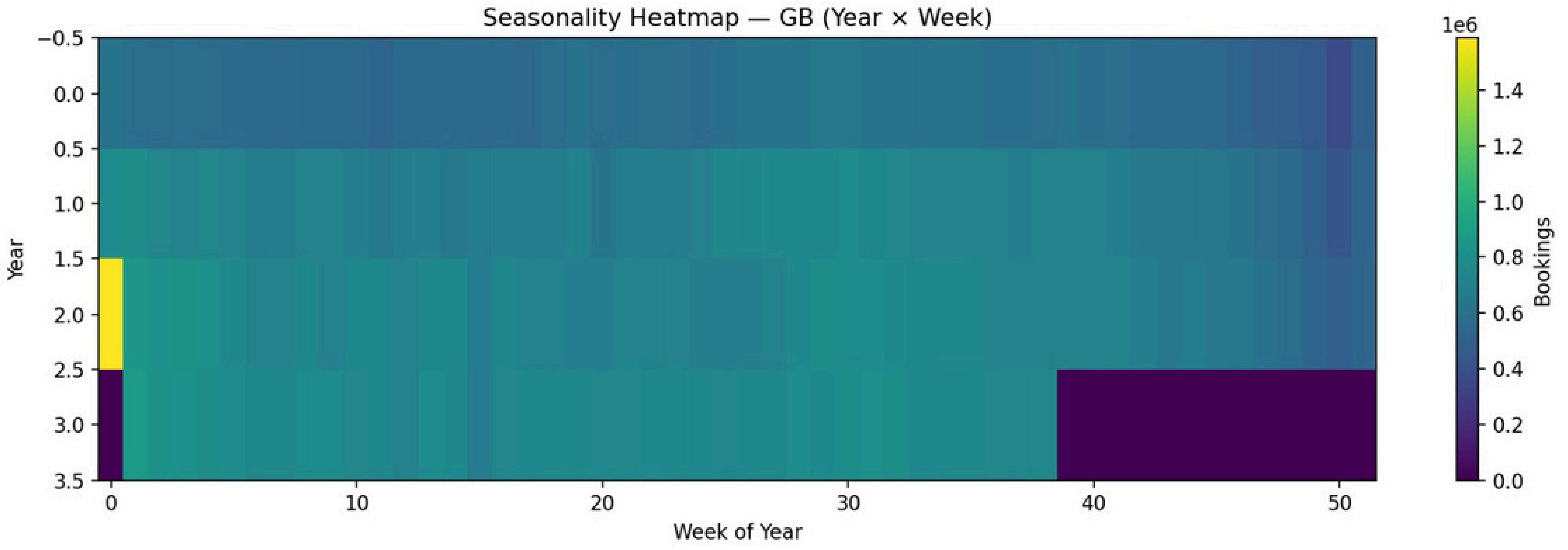
Country-Level Data

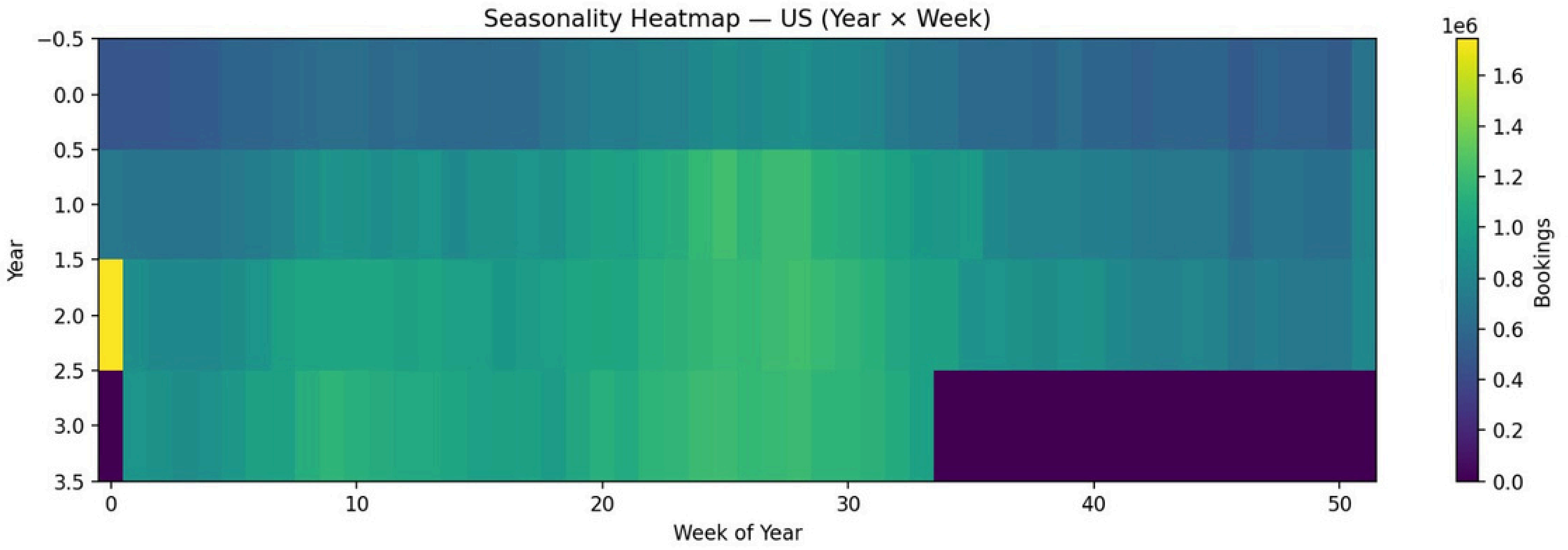
United States:
Strong, recurring peaks in summer (weeks 25–35) and holiday periods (weeks 45–52) reflect pronounced seasonality. Baseline activity remains steady year-round, aligning closely with marketing cycles.

United Kingdom:
Moderate seasonality with visible mid-summer and year-end lifts. Demand is steadier overall, suggesting resilience to seasonal dips but less responsiveness to marketing spikes.

Germany:
Clear summer surge with steeper post-peak declines than other markets, implying shorter booking windows and greater dependence on travel-season promotions.







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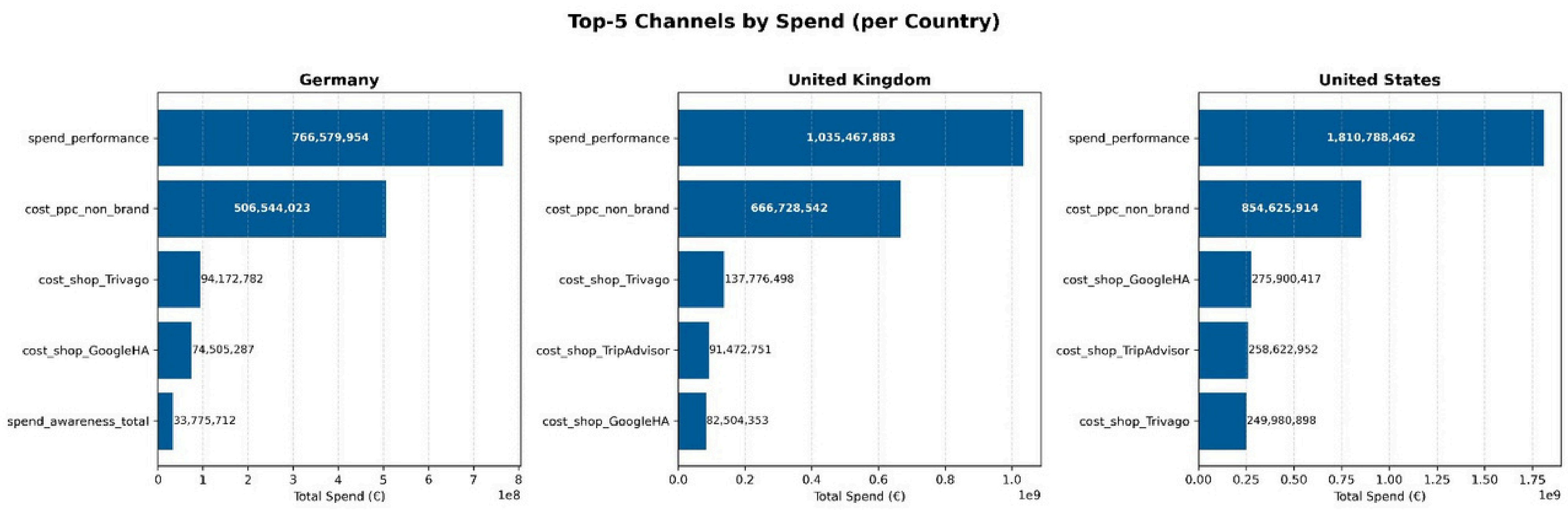
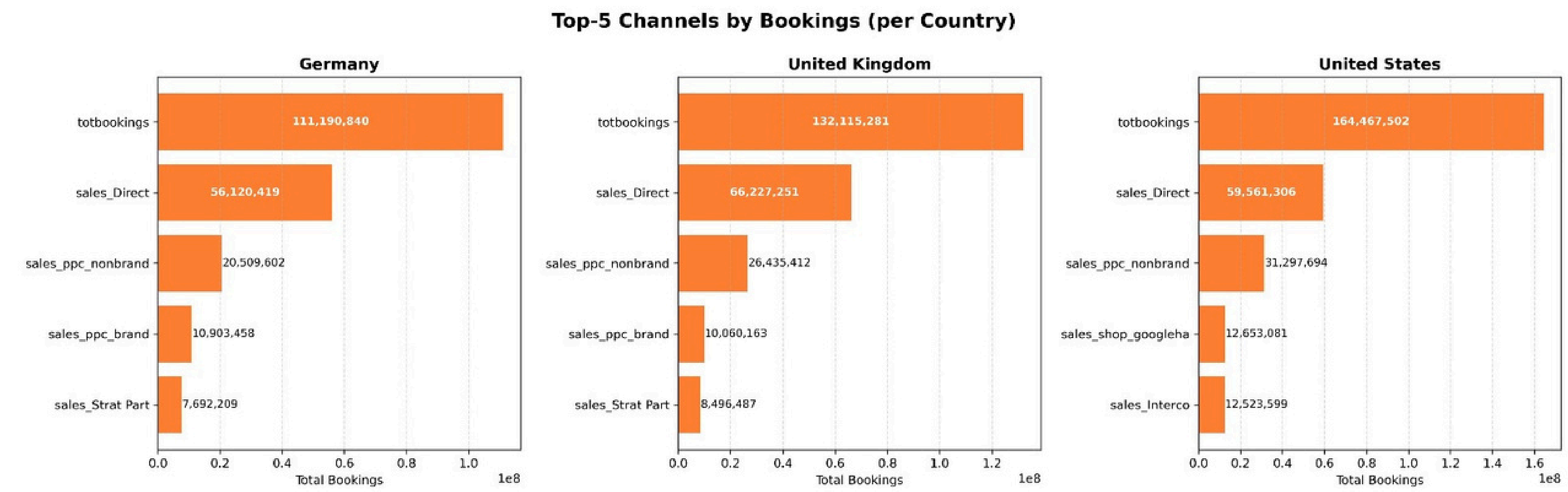
Country-Level Data





- **US:** Largest market – high spend and booking volume, led by performance channels.
- **UK:** Balanced spend mix, steady conversion across both awareness and performance.
- **Germany:** Leaner mix, heavier awareness share; lower spend but efficient ROI.
- **Overall:** Spend aligns with bookings – confirming performance-driven growth across all markets.

Appendix

Model Validation

Core Model - OLS

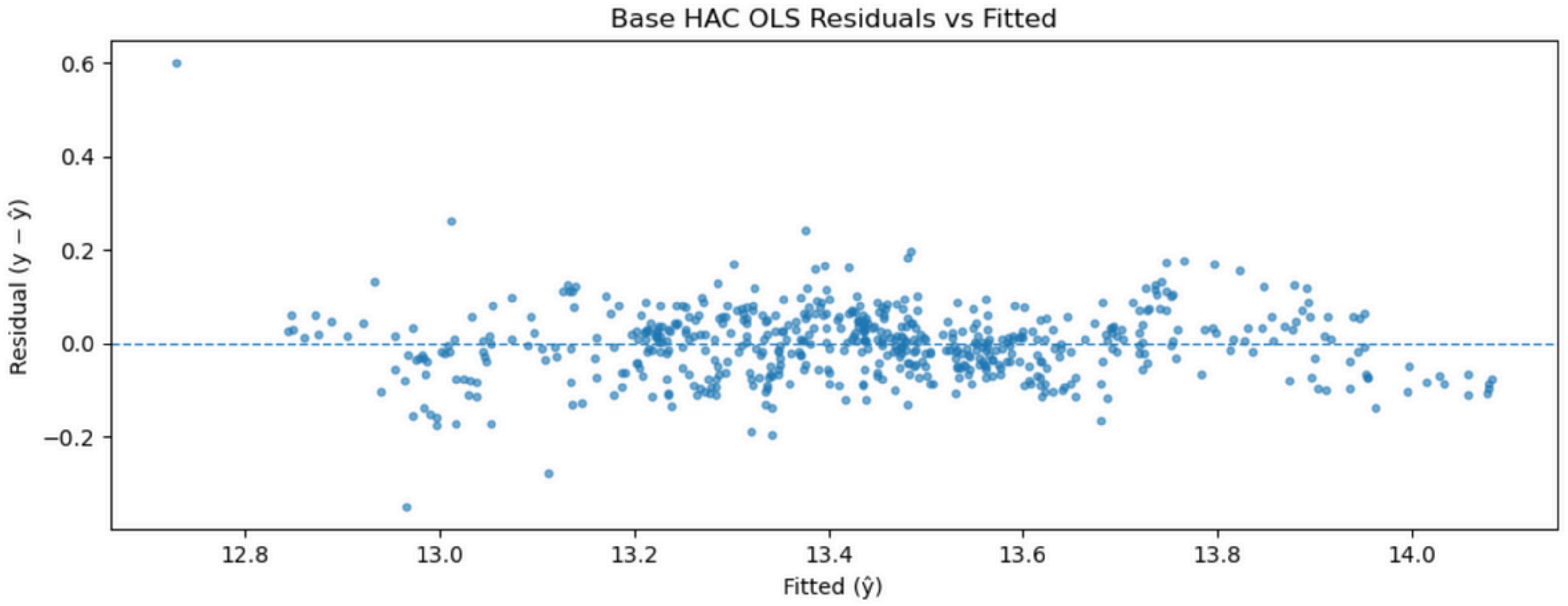
Log-Log with Adstock, Trend, Seasonality, Country Dummies

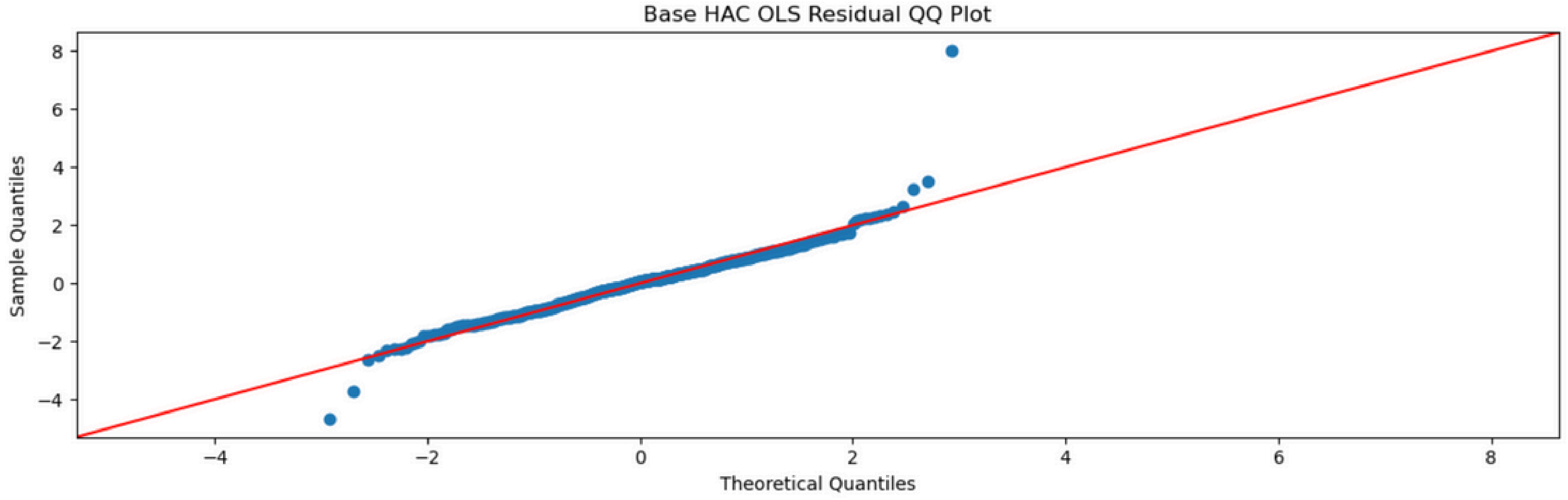
OLS Regression Results

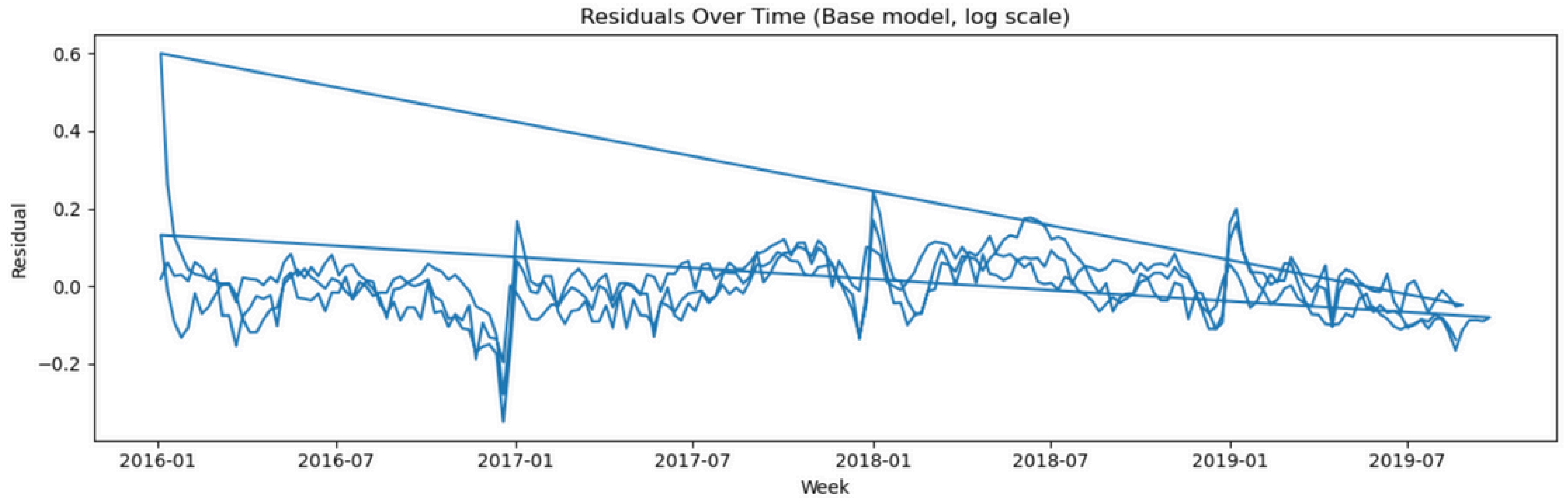
Dep. Variable:	y	R-squared:	0.918
Model:	OLS	Adj. R-squared:	0.916
Method:	Least Squares	F-statistic:	145.5
Date:	Sun, 02 Nov 2025	Prob (F-statistic):	1.09e-156
Time:	01:24:25	Log-Likelihood:	674.92
No. Observations:	576	AIC:	-1326.
Df Residuals:	564	BIC:	-1274.
Df Model:	11		
Covariance Type:	HAC		

	coef	std err	z	P>|z|	[0.025	0.975]
const	13.1684	0.018	739.290	0.000	13.133	13.203
f_perf	0.5922	0.083	7.168	0.000	0.430	0.754
f_aware	0.0040	0.001	2.805	0.005	0.001	0.007
f_brand	0.0040	0.001	2.805	0.005	0.001	0.007
t	0.0025	0.000	16.915	0.000	0.002	0.003
sin_woy	0.0445	0.010	4.448	0.000	0.025	0.064
cos_woy	-0.0986	0.012	-8.090	0.000	-0.122	-0.075
cty_de	-0.1622	0.015	-10.560	0.000	-0.192	-0.132
cty_us	0.2238	0.016	13.585	0.000	0.192	0.256
value (currency rate)_c	-0.0590	0.280	-0.211	0.833	-0.607	0.489
f_perf:cty_de	0.0044	0.090	0.048	0.961	-0.173	0.182
f_perf:cty_us	-0.1042	0.106	-0.983	0.326	-0.312	0.104
f_perf_x_brand	0.0557	0.006	8.663	0.000	0.043	0.068

Omnibus:	137.360	Durbin-Watson:	0.601
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1432.630
Skew:	0.722	Prob(JB):	8.10e-312
Kurtosis:	10.590	Cond. No.	2.16e+17







Appendix

Model Validation

AR(1) Corrected Model

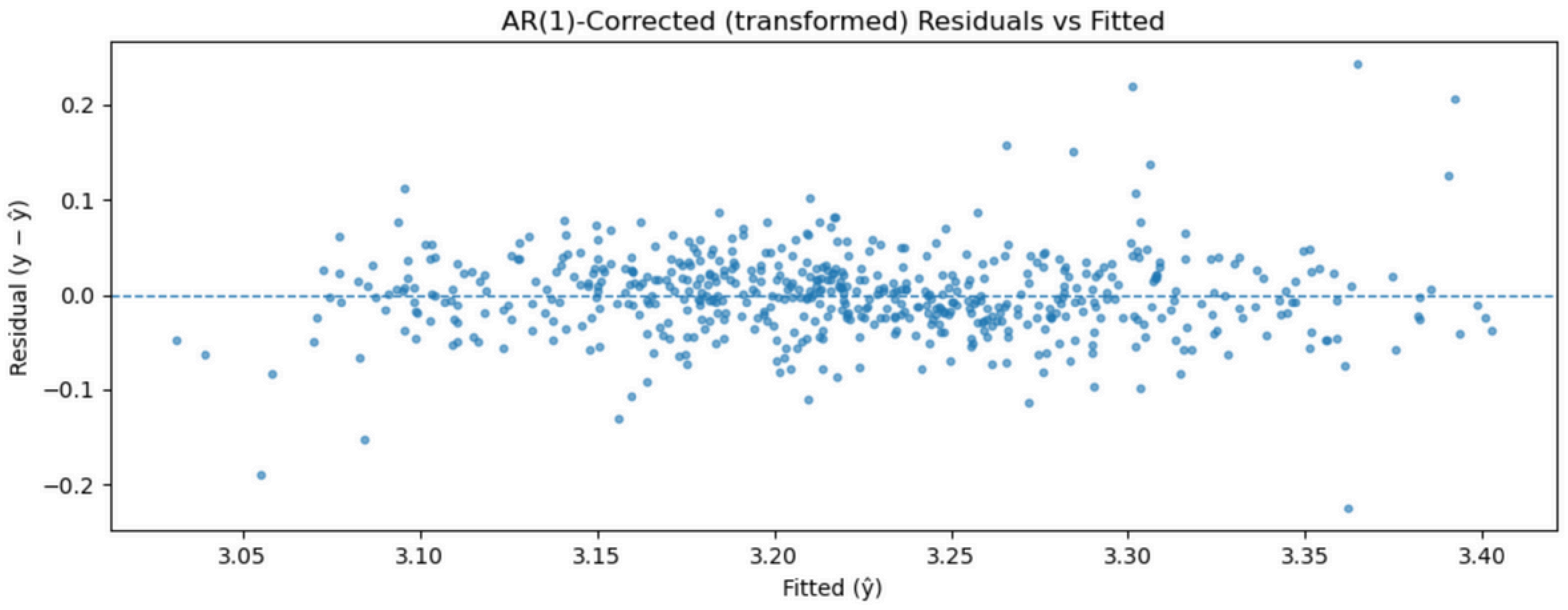
Validation for Serial Correlation

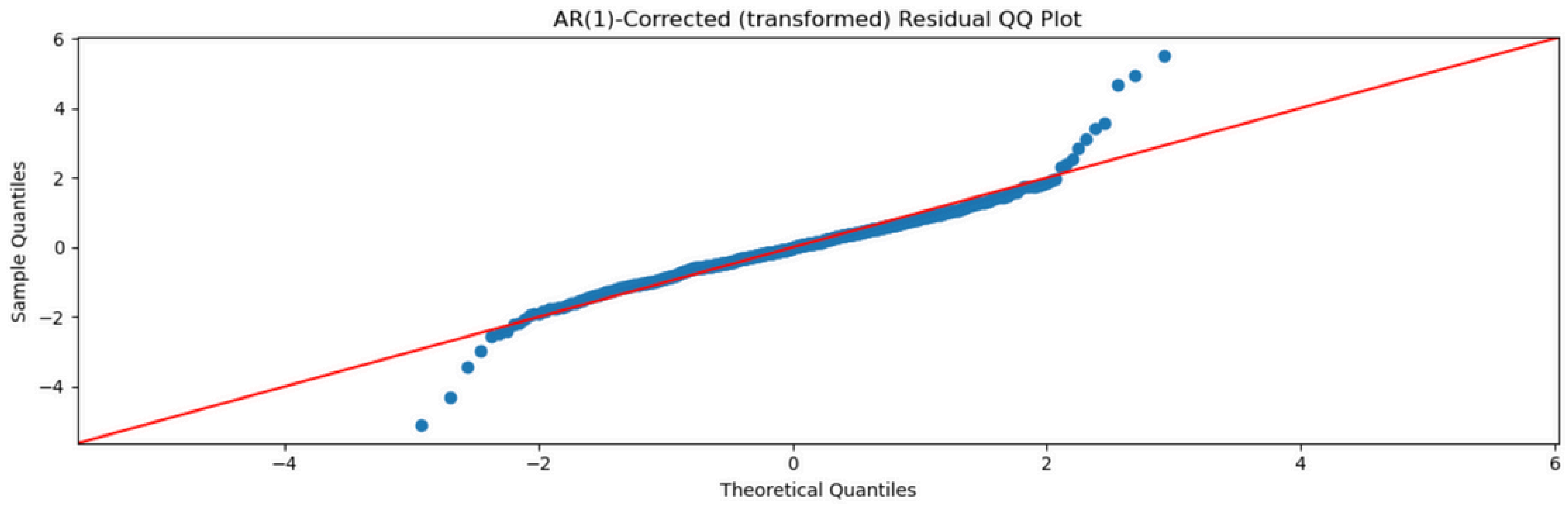
OLS Regression Results

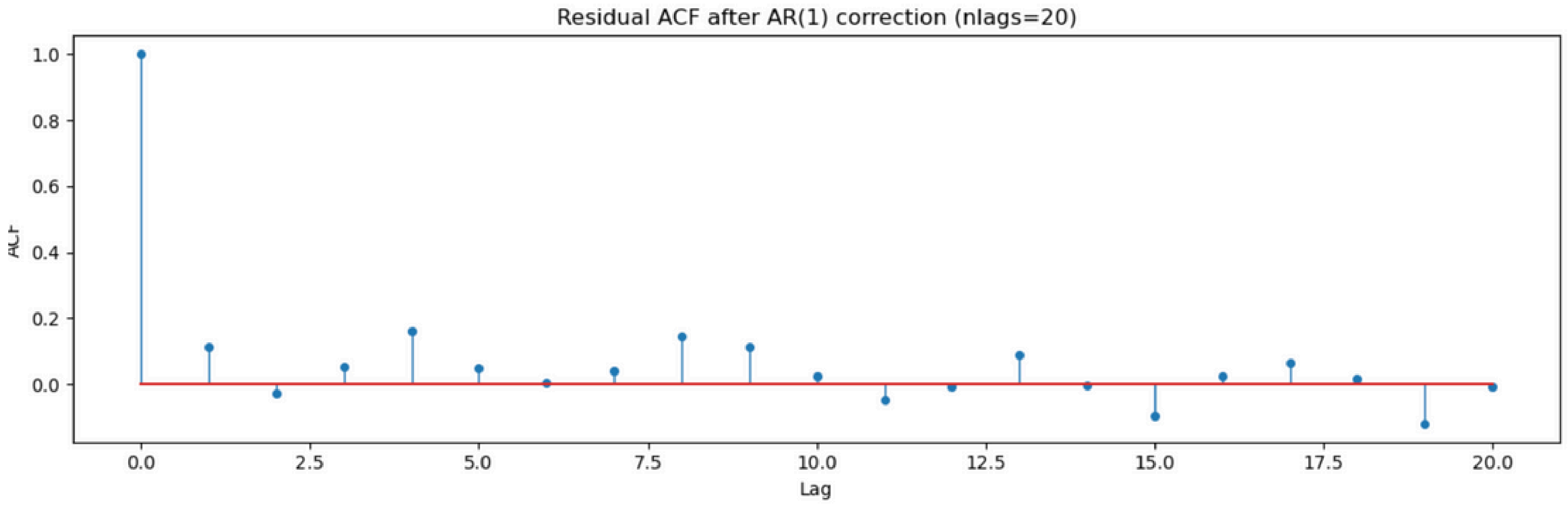
Dep. Variable:	y_star	R-squared:	0.726
Model:	OLS	Adj. R-squared:	0.721
Method:	Least Squares	F-statistic:	79.30
Date:	Sun, 02 Nov 2025	Prob (F-statistic):	1.19e-106
Time:	01:24:25	Log-Likelihood:	974.07
No. Observations:	573	AIC:	-1924.
Df Residuals:	561	BIC:	-1872.
Df Model:	11		
Covariance Type:	HAC		

	coef	std err	z	P>|z|	[0.025	0.975]
const	3.1466	0.005	588.867	0.000	3.136	3.157
f_perf	0.7670	0.151	5.075	0.000	0.471	1.063
f_aware	0.0022	0.001	2.616	0.009	0.001	0.004
f_brand	0.0022	0.001	2.616	0.009	0.001	0.004
t	0.0026	0.000	13.373	0.000	0.002	0.003
sin_woy	0.0491	0.011	4.417	0.000	0.027	0.071
cos_woy	-0.1217	0.013	-9.034	0.000	-0.148	-0.095
cty_de	-0.1480	0.023	-6.463	0.000	-0.193	-0.103
cty_us	0.2337	0.021	11.056	0.000	0.192	0.275
value (currency rate)_c	-0.4256	0.311	-1.369	0.171	-1.035	0.184
f_perf:cty_de	0.1084	0.172	0.631	0.528	-0.229	0.445
f_perf:cty_us	-0.2008	0.176	-1.138	0.255	-0.547	0.145
f_perf_x_brand	0.0659	0.010	6.357	0.000	0.046	0.086

Omnibus:	81.008	Durbin-Watson:	1.771
Prob(Omnibus):	0.000	Jarque-Bera (JB):	584.714
Skew:	0.351	Prob(JB):	1.07e-127
Kurtosis:	7.899	Cond. No.	6.19e+16





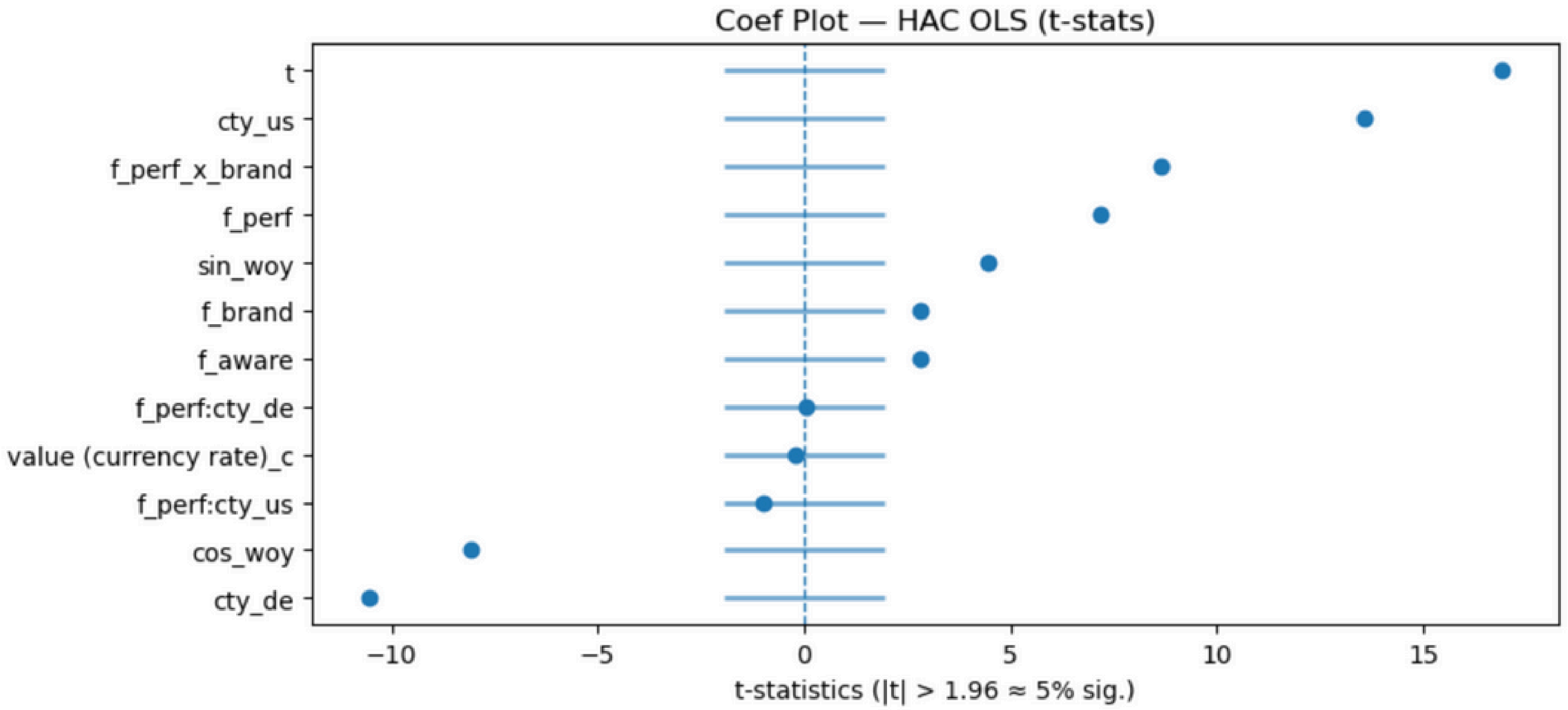


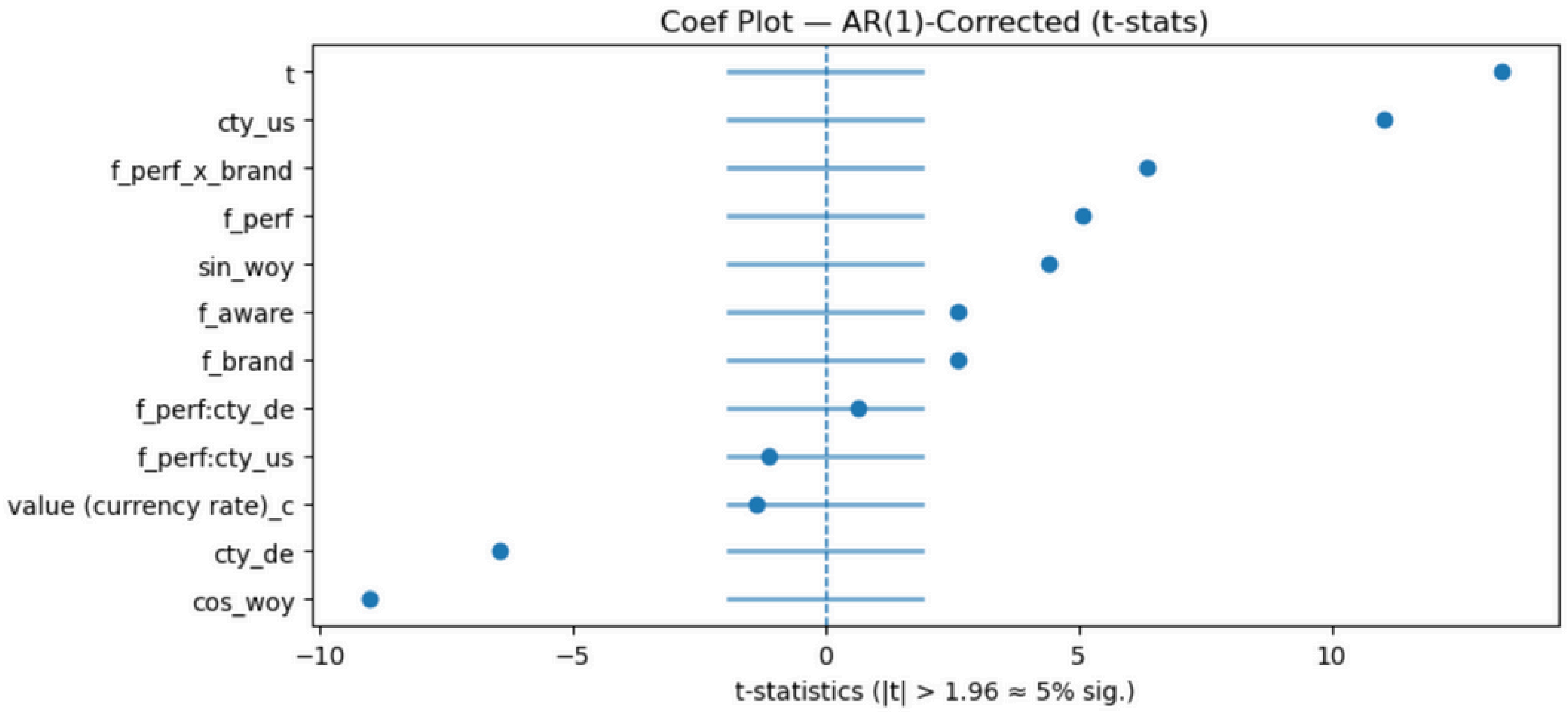
Appendix

Model Validation

Coef Plots

- Both models show consistent significance and direction for key predictors.
- Performance spend, trend, and seasonal terms are strong positive contributors, while Germany and cosine seasonality capture structural and cyclical effects.
- AR(1) correction slightly lowers t-values but preserves significance, confirming model stability.





Channel Effectiveness

Elasticities and Marginal Impact

- Performance channels demonstrate a high elasticity of ~0.77, meaning a 1% increase in spend drives ~0.77% more bookings (~5.4K additional weekly bookings).
- Awareness elasticity is near zero (0.002), confirming its limited short-run impact and role in long-term brand reinforcement.
- Across markets, responsiveness is strongest in the U.S., followed by Germany and the U.K., reflecting differences in signal strength and market maturity.

```
Elasticities at Mean (de-nested, AR(1)-corrected):
```

	channel_group	elasticity_at_mean
0	Performance	0.766998
1	Awareness	0.002213

```
[Story] ROMI panel shown as a relative index (set VALUE_PER_BOOKING for € impacts).

Elasticity → marginal impact panel:
```

	channel_group	elasticity_at_mean	Δbookings_per_+1%_spend	Δvalue_per_+1%_spend
0	Performance	0.766998	5429.886501	NaN
1	Awareness	0.002213	15.668922	NaN

```
[Elasticities] Performance elasticity ≈ 0.77 (1% spend → ~0.77% bookings). Awareness elasticity ≈ 0.00 short-run.
channel_group  elasticity_at_mean  Δbookings_per_+1%_spend  Δvalue_per_+1%_spend
  Performance            0.766998              5429.886501                   NaN
    Awareness            0.002213                15.668922                   NaN
Contribution (log scale): ['f_perf', 'f_aware']
```

	log_contrib	share_of_modeled_log_y
f_perf	-1.135385e-16	-8.450322e-18
f_aware	-1.501665e-18	-1.117644e-19

```
[Story] Country nuance: how strong are the average signals we're feeding into each market?
```

country	f_perf	f_aware
de	16.074080	12.051740
gb	16.359702	13.130538
us	16.909913	14.323637

Attribution & Scenarios

```
[Scenario snapshot]
Performance +10%  → 5.48% avg change in bookings
Awareness  +10%   → 0.05% avg change in bookings
+10% Branded PPC, −10% TV → −0.44% avg change in bookings
Guidance:
• Orthogonalized build lowers VIF while preserving the f_perf signal.
• Tell the short-run story with f_perf; track awareness on slower KPIs if you want long-run lift.
• If you want even tighter inference, ridge or drop the few higher-VIF controls and add one more seasonal pair.
```

Scenario Snapshot

- +10% Performance → +5.5% bookings
- +10% Awareness → +0.05% bookings
- +10% PPC / −10% TV → −0.4% bookings
- Performance drives ROI; awareness builds brand

Sensitivity sweep (expect near linear at 1–5%, some concavity by 20%):

	spend_shock	Δ% bookings (perf)	elasticity_perf	Δ% bookings (aware)	elasticity_aware
0	0.01	0.005582	0.558178	0.000050	0.004977
1	0.05	0.027684	0.553682	0.000244	0.004872
2	0.10	0.054829	0.548286	0.000475	0.004750
3	0.20	0.107634	0.538169	0.000905	0.004526

Sensitivity & Country Signals

- Near-linear response; mild concavity.
- Elasticity: perf ~0.6–0.8, aware ~0.0
- US strongest, UK moderate, DE lower
- ROI gains from smarter reallocation, not higher spend

Signals by country (means of transformed drivers):

	country	mean_f_perf	mean_f_aware
2	us	16.909913	14.323637
1	gb	16.359702	13.130538
0	de	16.074080	12.051740

Business Impact Calculations

ROI / Efficiency Approach

If total spend stays flat but marketing efficiency improves by +0.4% bookings, GlobeStay can either enjoy more bookings for the same spend, or spend less for the same bookings.

Savings≈Total Spend×0.4%

≈€984M×0.004=€3.94M

→ ≈ €4 million annual efficiency gains.

```
Annual Total Marketing Spend (in € Billions):
year
2016.0    0.948
2017.0    1.139
2018.0    1.028
2019.0    0.820
Name: total_spend, dtype: float64

Average Annual Spend (2016–2019): ≈ €0.984 B
```

Metric	Value	Source
Total annual marketing spend	**≈ €984 M**	2016–2019 average from spend totals
Performance share	**91.8% (€903 M)**	Performance spend share
Awareness (Digital) share	**1.8% (€18 M)**	Digital awareness share
Offline share	**6.4% (€63 M)**	Offline spend share
Proposed reallocation	**–5% from Offline → Performance**	Strategy proposal
Bookings uplift from reallocation	**≈ +0.4%**	Model-based scenario result